

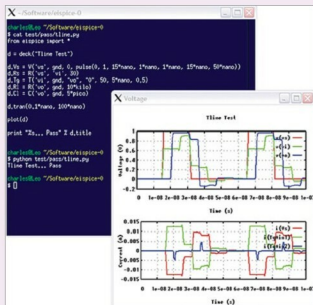


Fig. 2: eispace software

ter inhouse, so as to alleviate that concern. eispace lets you plot characteristic graphs, input and output by means of transient analysis and operating-point analysis.

In addition, a bevy of models also helps in increasing the usefulness of the program. These models are not part of a secret covenant, but are non-linear current and voltage source models, independent source models (like current generator/voltage generator), device models like resistance model, capacitance model or inductance model. Long-length transmission models are also incorporated into eispace by way of T lossless transmission model. There is also a diode model for diode circuits.

eispace provides access to Python language in order to manipulate SPICE decks and results. This feature lets you present the characteristics and plots as you desire, keeping in mind the customisability/readability of said project. The software has IBIS model import tool, plus all requisite drivers, receivers and pin models for easy use.

There is a new piece-wise cubic-spline extension for the incumbent

pwl, except for the addition of a pwl keyboard in order to spice it up. Models for VI curves and TA x VI curves have been improvised to increase clarity associated with these. Then there is an improved B-element that includes a time variable and is based on Python call-back, that is, it can use Python statements within B-element model.

Installation

eispace is available for free for Windows and Linux operating systems. Whereas for Windows you only need to install it like any other software, for Linux code you need to compile the code before you run it.

So for win edition, download and install Windows version of Python 2.5 and then download and install Numpy, before finally downloading and installing eispace. As of time of writing, in order to work with eispace version 0.9, you need to install the additional Python and Numpy Python Module installed on your system.

Running eispace

To fire up Python, run Python from the command line and then import eispace module with Import eispace command. Then, check out Python scripts in the test/pass directory for some examples, or try out the example below:

```
import eispace
cct = eispace.Circuit("Capacitor Test")
cct.Vx = eispace.V(1, eispace.GND, 4,
eispace.Pulse(4, 8, '10n', '2n', '3n',
'5n', '20n'))
cct.Cx = eispace.C(1, eispace.GND,
'10n')
```

```
cct.tran('0.5n', '100n')
eispace.plot(cct)
```

Just like SPICE 3, eispace statements also come in three distinct varieties: component definition (these are device names, types, connection nodes and parameter values), analysis commands (these obviously work on the analysis part, beefing up the back-end analyses) and post-processing commands (like plot, these give a visual look to the data that you get otherwise).

The direction the future is staring at

The primary new feature for the next release is supposed to be this revolutionary W-element-like multi-conductor, frequency-dependent model. Plans for eispace also reveal something known as a digital signalling layer, which assists in driving real signals and not just the rising or falling edge.

To add to this list, a graphical schematic entry tool and an even better graphing tool is expected to replace the obsolete counterparts as of now.

There are many more features that have been promised by the developers of eispace project. They say, you can get the hang of Python from a newbie level to beginner level in a mere half an hour. Though the time seems adequate for Mensa people, even I was able to do some Python coding after only a couple of hours!

Charles Eidsness is the main developer behind eispace, and it has been nothing short of a miracle for many simulation users who wanted to use a front-end driven neatly by Python. So go check out the DVD, provided with this month's *Electronics for You Plus*, install eispace and find out first-hand what we are talking about. **EFY**

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